

STAT 801A Syllabus (Fall 2026)

Statistical Methods In Research – Non Calculus

Instructor

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All office hours are guaranteed for the first 30 minutes. If no students are present after 30 minutes, office hours are ended.

Course Description

Prerequisites: Introductory course in statistics.

Statistical concepts and statistical methodology useful in descriptive, experimental, and analytical study of biological and other natural phenomena. Practical application of statistics rather than on statistical theory.

Statistical Software

The statistical computing software package R will be used extensively to perform calculations in this class. Additionally, we will make use of RStudio (GUI for R) and Quarto (document creation).

1. Download R: <https://cran.r-project.org>
2. Download RStudio: <https://posit.co/downloads>
3. Download Quarto: <https://quarto.org/docs/download/>

Assessments

This course consists of quizzes, projects, and exams.

Projects/Quizzes: Projects and quizzes will be assigned periodically throughout the semester. You will have at least one week to complete the assessment.

Midterm: There will be one midterm approximately halfway through the semester. Details will be provided in class.

Final: To be announced in class.

These assessments are weighted as follows:

	Projects/quizzes*	Midterm	Final
% of grade	20%	30%	50%

*All projects need to be turned electronically via PDF documents. A project completed in an unreadable or unprofessional manner will be returned for a zero grade. Quizzes will be assigned on Canvas. No late projects or quizzes are accepted.

If projects allow for group work, all group members are expected to participate equally and have a complete understanding of all components for it. I will lower a student's project grade if he/she/they do not abide by this group work policy.

Grades

Your overall course grade percentage guarantees the following grade for the course.

Letter Grade	+	-
A	93	90
B	87	83
C	77	73
D	67	63
F	<60	

Continuity of Instruction Policy

In the event that the university is closed during our regular class hours, please check the Canvas announcements for alternative class arrangements.

AI Disclaimer

The recent popularity of large language models and other artificial intelligence tools introduces a unique challenge in the academic community. As an instructor, my goal is to promote critical thinking for the topics taught in this course. Unless otherwise stated, **the use of AI tools is not permitted**. If you are suspected of using AI, I reserve the right to give a zero grade.

Tentative Topics

- Intro to R
- Data summaries
- Probability
- Inference for one mean
- Inference for two means
- Inference for proportions
- Inference for variances
- ANOVA
- Regression

Important Dates

- No lab on 8/24
- No lab on 9/7 (Labor Day)
- No class on 10/19 (Fall Break)
- No class or lab on the week of 11/23
- Final: Monday, Dec. 14 from 10:00 to noon

UNL Course Policies and Resources

Students are responsible for knowing the university policies and resources found on this page:

<https://go.unl.edu/coursepolicies>

- University-wide Attendance Policy
- Academic Honesty Policy
- Services for Students with Disabilities
- Mental Health and Well-Being Resources
- Final Exam Schedule
- Fifteenth Week Policy
- Emergency Procedures
- Diversity & Inclusiveness
- Title IX Policy
- Other Relevant University-Wide Policies